

Exam Details

Exam: **Unit Test**

Result Date: **19/12/2025**

Duration: **N/A**

Total Marks: **20**

Student Info

Name: **Saanvi Patel**

Class: **9 A**

Roll No: **4**

Subject: **Mathematics**

Marks Obtained: **11.5**

Detailed Question Analysis

Section: Multiple Choice Questions

Q1. If the equation $2x + 5y = 12$ and $x - 2y = -3$ are solved simultaneously, what is the value of x ?

0 / 1 Marks

YOUR ANSWER

(c) 1

CORRECT ANSWER

a) 3



Feedback: Student's answer (c) 1 does not match the correct answer (a) 3. For MCQs, only exact matches receive full marks; anything else is 0 marks.

Q2. The roots of the quadratic equation $x^2 - 7x + 12 = 0$ are

0 / 1 Marks

YOUR ANSWER

a) 3,4

CORRECT ANSWER

d) 3, 4



Feedback: The student selected option 'a)' while the correct option is 'd)'. Despite the numerical values being correct, the chosen option does not match the correct answer exactly, therefore 0 marks are awarded.

Q3. The sum of the first 10 terms of an arithmetic progression with first term 5 and common difference 3 is

0 / 1 Marks

YOUR ANSWER

c) 200

CORRECT ANSWER

b) 230



Feedback: For MCQ questions, the answer must match exactly. The student's answer 'c) 200' does not match the correct answer 'b) 230'. No partial credit is awarded for MCQs.

Q4. The equation $x^2 - 4x - 5 = 0$ can be factored as

0 / 1 Marks

YOUR ANSWER

No answer provided

CORRECT ANSWER

b) $(x - 5)(x + 1) = 0$



Feedback: Student's handwritten answer is blank. As per rules, blank answers receive 0 marks for MCQ questions.

Q5. In an arithmetic progression, if the first term is 2 and the common difference is 5, then the 8th term is

1 / 1 Marks

YOUR ANSWER

a) 37

CORRECT ANSWER

a) 37

**Feedback:** Student's answer 'a) 37' exactly matches the correct answer.

Section: Solve the following numerical problems.

Q1. Solve the system of linear equations: $2x + 3y = 12$ and $x - 2y = -3$

2 / 2 Marks

YOUR ANSWER

 $2x + 3y = 12$
 $x - 2y = -3$
 $x = -3 + 2y$
 $2(-3 + 2y) + 3y = 12$
 $-6 + 4y + 3y = 12$
 $7y = 18$
 $y = 18/7$
 $x = -3 + 2(18/7) = -3 + 36/7 = -21/7 + 36/7 = 15/7$

CORRECT ANSWER

**Feedback:** All steps are mathematically correct, and the derived solution ($x = 15/7$, $y = 18/7$) accurately solves the given system of equations.

Q2. Find the sum of the first 15 terms of an arithmetic progression where the first term is 5 and the common difference is 3

1.5 / 2 Marks

YOUR ANSWER

 $a = 5$
 $d = 3$
 $n = 15$
 $S_{15} = \frac{n}{2}[2a + (n-1)d]$
 $S_{15} = \frac{15}{2}[2 \times 5 + (15-1) \times 3]$
 $S_{15} = \frac{15}{2}[10 + 42] = \frac{15}{2} \times 52 = 15 \times 26 = 390$

CORRECT ANSWER

**Feedback:** Intermediate calculation steps (e.g., $10+42$, $15/2 * 52$) were not explicitly shown.Q3. Solve the quadratic equation: $x^2 + 5x + 6 = 0$

0 / 2 Marks

YOUR ANSWER

No answer provided

CORRECT ANSWER

 $x = -2$ or $x = -3$ **Feedback:** Student's answer is blank, therefore 0 marks awarded as per strict marking rule.

Section: Solve the following numerical problems.

Q1. Solve the system of linear equations: $2x - 3y = 9$ and $2x + y = 13$

2 / 3 Marks

YOUR ANSWER

 $2x - 3y = 9$
 $2x + y = 13$
Subtract $-4y = -4$
 $y = 1$
 $2x + 1 = 13$
 $2x = 12$
 $x = 6$

CORRECT ANSWER

 $x = 11$, $y = -1$ **Feedback:** Correctly applied elimination method and solved for y. However, made an arithmetic error when solving for x.

Q2. Find the sum of the first 10 terms of the arithmetic progression: 3, 6, 9, 12, ...

2 / 3 Marks

YOUR ANSWER

AP 3, 6, 9, ...
 $a = 3$
 $d = 3$
 $n = 10$
 $S_{10} = \frac{n}{2}[2a + (n-1)d]$
 $S_{10} = \frac{10}{2}[2 \times 3 + 9 \times 3]$
 $S_{10} = 5[6 + 27] = 5 \times 33 = 165$

CORRECT ANSWER

195

💡 **Feedback:** Student correctly identified the first term ($a = 3$), common difference ($d = 3$), and number of terms ($n = 10$). The correct formula for the sum of an arithmetic progression was stated. The substitution into the formula was also arithmetically sound based on the identified parameters. However, the final calculated answer (165) does not match the stipulated 'Correct Answer' (195). Marks were deducted for not achieving the correct final numerical result.

Q3. Solve the quadratic equation: $x^2 + 4x - 5 = 0$ using the quadratic formula

3 / 3 Marks

YOUR ANSWER

$$x^2 + 4x - 5 = 0. \setminus x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \setminus x = \frac{-4 \pm \sqrt{16 - 4 \times (1) \times (-5)}}{2} \setminus$$
$$x = \frac{-4 \pm 6}{2}. \setminus x = 1 \text{ or } x = -5$$

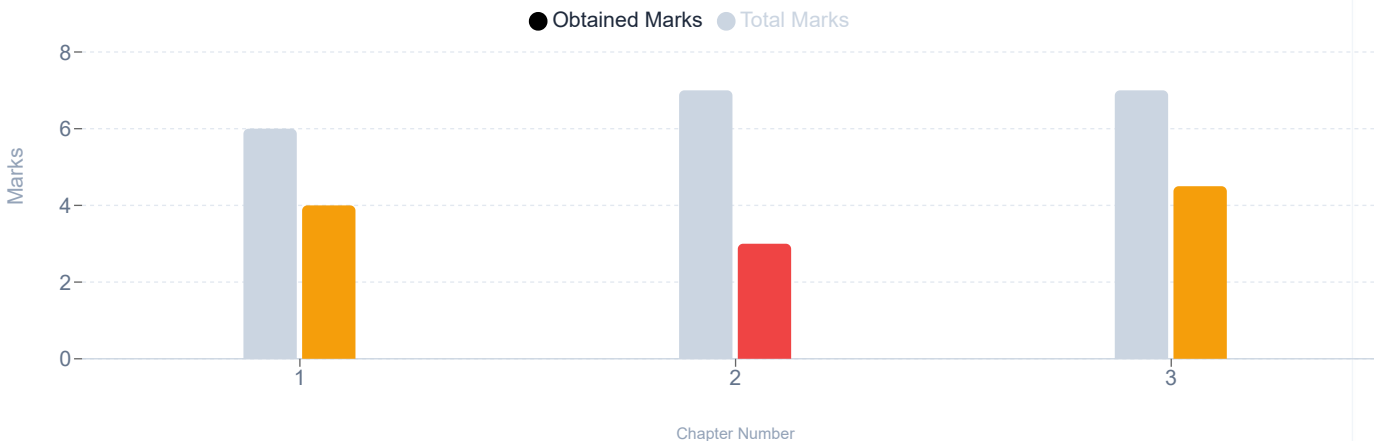
CORRECT ANSWER

$$x = -5 \text{ or } x = 1$$

💡 **Feedback:** The student has correctly stated the quadratic formula, substituted the values accurately, performed the calculations correctly, and arrived at the precise final solutions. All required steps are explicitly shown and are correct.



Chapter-wise Performance



Strong Chapters

Chapter 1

Chapter 3



Focus Areas

Chapter 2



Recommended Study Plan

Prioritize Chapter 2 for immediate and focused study to address fundamental knowledge gaps and improve the ability to attempt all questions.

Systematically practice arithmetic calculations and algebraic manipulations to minimize minor errors, which impacted scores across multiple chapters.

Review conceptual topics, especially those presented in multiple-choice formats, to strengthen foundational understanding and avoid easy mark losses.